

Neurological Representations of Linguistic Intuitions and World Knowledge Intuitions: a Case Study of Evidentiality

Yurie Hara

1 Introduction

This chapter aims to dissociate linguistic intuitions and world knowledge intuitions by re-examining the introspection-based judgments of Japanese modality and evidentiality reported in Hara & Davis (2013); Davis & Hara (2014) and Hara (2017) as well as the results of an electroencephalography (EEG) experiment conducted by Hara, Orita, Ying, Koshizuka & Sakai (2020).¹

In the psycho/neuro-linguistics literature, several kinds of semantic/pragmatic violation are often conflated into world knowledge violations. Specifically, in ERP studies, starting from the seminal work by Kutas & Hillyard (1980) anomalous sentences such as (1b) are often considered semantically/pragmatically ill-formed and elicit an ERP component called N400 (negative deflection that peaks around 400 milliseconds).

- (1) a. Control: He spread the warm bread with butter.
- b. Semantic Violation: He spread the warm bread with socks.

Pylkkänen, Oliveri & Smart (2009) point out, however, that anomalous sentences such as (1b) or *Dutch trains are sour* are actually semantically well-formed but violating world knowledge (See Section 2.3). Pylkkänen et al. (2009) employ magnetoencephalography (MEG) to identify neural responses to semantic violations that cannot be reduced to world knowledge violation, and found an effect in the Anterior Midline Field (AMF), localised to the ventromedial prefrontal cortex (vmPFC). This result identifies the AMF component as a potential neural response specific to linguistic semantic composition. Pylkkänen et al. (2009) also note that the vmPFC may likewise be sensitive to another domain of the meaning comprehension, i.e., pragmatic phenomena, such as implicatures or presupposition failure, since deficit-lesion studies have demonstrated that damage to the ventromedial region results in significant impairments in understanding sarcasm (Shamay-Tsoory et al., 2005).

The goal of this chapter is to seek the neurological differences of the pragmatic and world knowledge intuitions and to identify the neural correlate of pragmatic processing parallel to the ones found by Pylkkänen et al. (2009) for semantic processing. Specifically, this chapter reexamines the results of EEG experiments of Japanese evidentiality and modality conducted by Hara et al. (2020).

The contrast between the Japanese evidential *yooda* and the modal *daroo* offers an optimal framework for pursuing the objectives of this chapter. According to Hara & Davis (2013); Davis & Hara (2014); Hara (2017, henceforce, H&D), asserting *p-yooda* presupposes a causal relation *p* causes *q* (e.g., rain causes wet streets) and asserts that the speaker perceives a *q* state (e.g., wet

¹EEG is a monitoring method to record event-related potentials (ERPs), which are electrical brain responses caused by a sensory, cognitive, or motor stimuli.

streets) as in (2a). (2b) is anomalous because the utterance presupposes an implausible causal relation, i.e., wet streets cause rain, thus it is an instance of a world knowledge violation.

- (2) a. (Looking at wet streets)
Ame-ga futta yooda.
rain-NOM fell EVID
‘It seems that it rained.’
- b. (Looking at falling raindrops)
#Michi-ga nureteiru yooda.
streets-NOM wet EVID
‘#It seems that the streets are wet.’

At the first glance, *daroo* seems to be in complementary distribution with *yooda* as can be seen in (3). However, a closer examination reveals that the reason why (3a) is anomalous is not because it presupposes an implausible causal relation, but because it fails to utter the optimal sentence in the situation, namely *p-yooda* as in (2a). In other words, (3a) is pragmatically blocked in the sense of Optimality Theory Pragmatics (Blutner & Zeevat, 2004), hence it is an instance of pragmatic violation. (3a) is not a semantic violation as it does not break any semantic composition rules, nor a world knowledge violation as inferring the high probability of rain based on the observation of wet streets is a very plausible situation.

- (3) a. (Looking at wet streets)
#Ame-ga futta daroo.
rain-NOM fell I-bet
‘#Probably, it rained.’
- b. (Looking at falling raindrops)
Michi-ga nureteiru daroo.
streets-NOM wet I-bet
‘Probably, the streets are wet.’

This chapter is organized as follows: Section 2 reviews a brief history of neurolinguistic research on meaning comprehension or “semantic” violations. In brief, the classic N400 ERP component is not a violation of semantic rules but world knowledge (Pykkänen et al., 2009) and both semantic/pragmatic and world knowledge violations have been associated with certain neural responses in both ERP and MEG research. Section 3 outlines the research questions of the current chapter: Can EEG dissociate world knowledge violations from purely linguistic violations? Is there an ERP component that is a correlate of pragmatic processing? Section 4 provides an overview of semantics and pragmatics of Japanese evidential *yooda* and modal *daroo* by reviewing Hara & Davis (2013); Davis & Hara (2014) and Hara (2017). Finally, Section 5 examines the results of the EEG study conducted by Hara et al. (2020) in order to assess whether the research questions outlined in Section 3 can be addressed. To anticipate the results, while the anomalous *yooda* sentences elicit N400 as expected, there was no significant difference between control and target *daroo* sentences. Thus, the study did find a difference between the processing of world knowledge and pragmatic violations, but the neural correlate of pragmatic anomaly was not identified. Notably, compared to *yooda* (both control and target) sentences, *daroo* sentences yield N400 effect that has been identified as a processing of modality, i.e., multiple possible worlds (Dwivedi et al., 2006; Kulakova et al., 2014; Kulakova & Nieuwland, 2016). This modality-induced N400 may mask or interfere with sustained frontal negative shift, which is alleged to be an ERP version the AMF MEG effect.

2 Neurological studies on (non-)linguistic intuitions

This section provides a brief overview of neurolinguistic research on “semantic” processing. The N400 effect was first discovered by Kutas & Hillyard (1980) (Section 2.1). In Section 2.2, we review Hagoort, Hald, Bastiaansen & Petersson (1998), who proposed a distinction between *semantic violations*—implausible or impossible situations—and *world knowledge violations*—statements that are false with respect to common knowledge—and demonstrated that both types of violations nonetheless elicit the same N400 response. Section 2.3 focuses on Pykkänen, Oliveri & Smart (2009), who argued that the “semantic violations” studied by Hagoort et al. (1998) in fact constitute world knowledge violations, since they do not infringe on the rules of semantic composition. Using MEG, they showed that genuine semantic violations evoke activation in the vmPFC (the AMF effect), whereas world knowledge violations are associated with activation in the left inferior prefrontal cortex (LIPC). Because the claim of Pykkänen et al. (2009) rests primarily on MEG evidence rather than EEG, Section 2.4 turns to ERP studies that have investigated the neural correlates of difficulties in discourse referent ambiguity (Nieuwland & Van Berkum, 2008; Van Berkum et al., 2003, 2007).

2.1 N400 as “semantic” violation

The classic work by Kutas & Hillyard (1980) is the first study that investigated ERPs elicited when an inappropriate word appears unexpectedly at the end of a sentence. In their experiments, participants read seven-word sentences with three different conditions as in (4). (4a) is a control sentence which is both syntactically and semantically well-formed. (4b) is a target sentence that ends with a semantically anomalous word. (4c) is a semantically appropriate sentence but the final word is presented with larger fonts.

- (4) a. Control: He spread the warm bread with butter.
 b. Semantic Violation: He spread the warm bread with socks.
 c. Large Font: He spared the warm bread with **butter**.

As depicted in Figure 1, the semantically anomalous seventh words elicited a negative ERP component (N400), which began around 250 ms after stimulus onset and peaked at approximately 400 ms. Thus, Kutas & Hillyard (1980) conclude that the “N400 wave may be an electrophysiological sign of the ‘reprocessing’ of semantically anomalous information” (203).

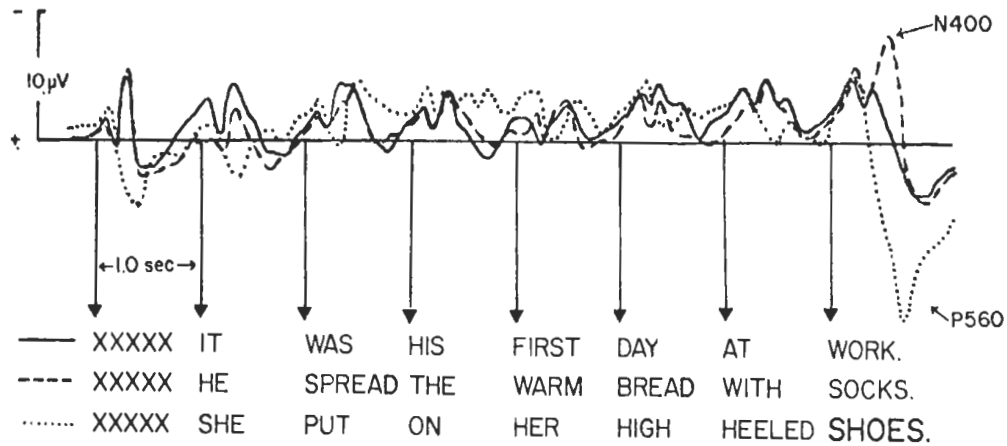


Figure 1: Kutas and Hillyard's (1980) N400 effect (203)

2.2 N400 as both “semantic” and world knowledge violation

Hagoort, Hald, Bastiaansen & Petersson (1998) critically review Kutas & Hillyard (1980) and point out that sentences can be anomalous in terms of their meanings in two ways. First, a sentence can be plausible, hence semantically well-formed, but false as in (5b) since all trains in Holland are yellow as correctly stated in (5a). Second, (5c) is anomalous as it violates the semantic rules of English. That is, the adjective *sour* has a linguistic/semantic requirement that it needs to be predicated with something edible, but *Dutch trains* do not meet this requirement.

- (5)
- a. control: Dutch trains are yellow.
 - b. world knowledge: Dutch trains are white.
 - c. Semantic violation: Dutch trains are sour.

As seen in Figure 2, both semantic and world knowledge violations elicited similar N400 effects, indicating that lexical semantics and world knowledge are integrated during the same time window, beginning around 300 ms after word onset.

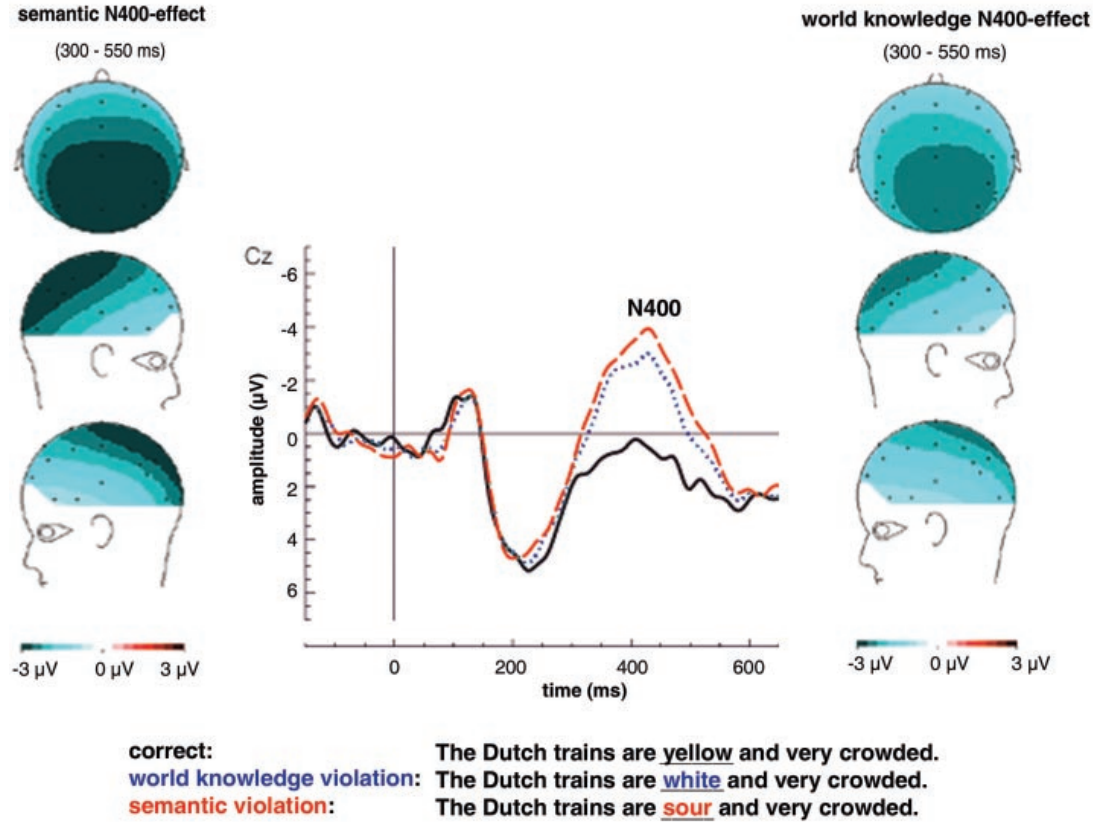


Figure 2: Hagoort et al.'s (1998) N400 effect (439)

2.3 AMF effect as purely linguistic semantic violation

As briefly discussed in Section 1, Pykkänen, Oliveri & Smart (2009) argue that the “semantic” violations assumed by Kutas & Hillyard (1980) and Hagoort et al. (1998) such as (1b) and (5c) are not purely linguistic/semantic violations but world knowledge violations because they do obey the semantic composition rules but denote implausible situations (Pykkänen & McElree, 2007). That is, in terms of semantic composition, both *Dutch trains* and *sour* are predicates that denote a property of type $\langle e, t \rangle$, i.e., a set of individuals, thus they can both serve as arguments of the generic operator G of type $\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle$. In terms of world knowledge, the sentence describes an impossible situation as trains are not edible so we cannot taste them, thus it is still a world knowledge violation.

In order to provide a direct comparison between semantic violations and world knowledge violations, Pykkänen et al. (2009) investigated the semantic constraints governing verbal *un*-prefixation in English. The use of the prefix *un*- is subject to specific morphological and semantic restrictions: In general, *un*-verbs are most naturally derived from stems that denote actions or processes which move an entity into a marked or non-default state (Covington, 1981). The derived *un*-verb then expresses the reversal of this action, namely a transition back to an unmarked, default, or entropic state (Horn, 2002).

On the basis of this constraint, examples such as (6b) are judged semantically ill-formed. Although the situation described is in principle possible and coherent, the verb *unchill* is anomalous because it does not encode a reversal to an unmarked or entropic state, and thus violates the semantic licensing conditions of *un*-prefixation. In contrast, examples such as (6c) fall into the category of

world knowledge violations, comparable to (4b) and (5c). These cases are semantically well-formed in that the prefixal derivation of the verb is fully licensed, but the scenario described is implausible or impossible given shared knowledge of the world.

- (6) a. Well-formed: ... the wine was being uncorked...
- b. Semantic violation: ... the wine was being unchilled ...
- c. World violation: ...the thirst was being uncorked...

Pylkkänen et al. (2009) employ magnetoencephalography (MEG) to investigate neural responses to semantic violations that cannot be reduced to instances of world knowledge violations.²

As shown in Figure 3, The vmPFC exhibited significantly greater activation for semantic violations relative to controls at 290–310 ms, indicating a semantic-specific effect consistent with its role in composition rather than real-world plausibility evaluation. Additionally, semantic violations elicited increased activity in the left inferior prefrontal cortex (LIPC) at 300–330 ms, and world knowledge violations produced a similar LIPC effect within the same time window. These findings replicate Hagoort et al.’s (1998) results and further demonstrate that the LIPC is sensitive to both linguistic semantic constraints and world knowledge, without distinguishing between them.

²Magnetoencephalography (MEG) is a neuroimaging technique that records the magnetic fields generated by neuronal currents. Like electroencephalography (EEG), it provides millisecond-level temporal resolution, but because magnetic fields are less distorted by the skull and scalp than electrical signals, MEG offers superior spatial localization of cortical activity.

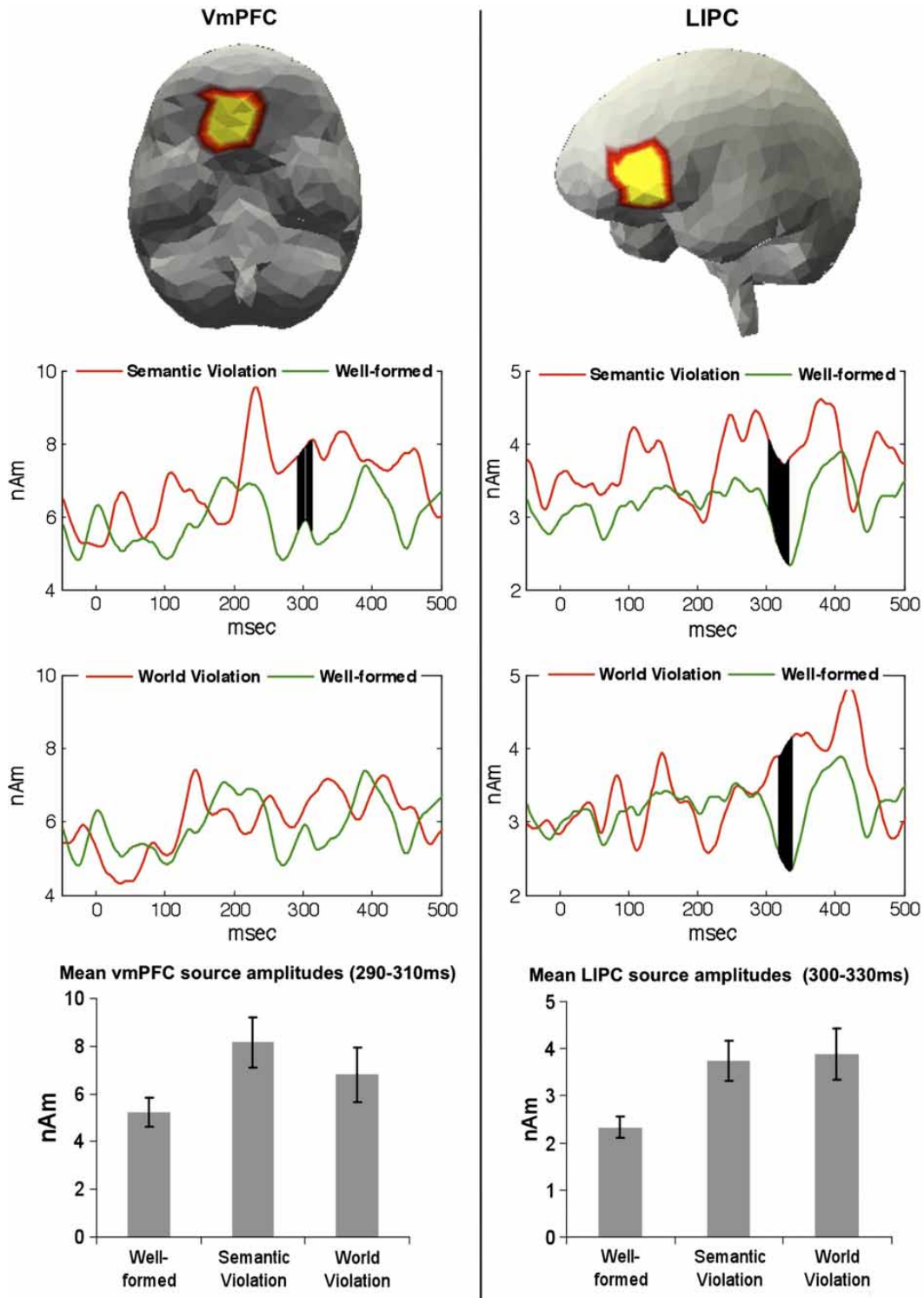


Figure 3: Pylkkaenen et al's (2009) results of the vmPFC and LIPC ROI analysis (1323)

Furthermore, Pylkkänen et al. (2009) observe that “one might expect the vmPFC to also be sensitive to other more classically ‘pragmatic’ phenomena, such as implicatures or presupposition

failure” (1332), given that deficit-lesion studies have shown that damage to the ventromedial region leads to significant impairments in comprehending sarcasm (Shamay-Tsoory et al., 2005).

While Pylkkänen et al. (2009) identified the AMF effect localized to vmPFC as a neurological correlate of semantic violation in their MEG study, it is yet to be investigated whether there is a comparable ERP component that is sensitive to semantic/linguistic violations. Pylkkänen et al. (2009) speculate that sustained anterior negativities observed by Nieuwland & Van Berkum (2008); Van Berkum, Brown, Hagoort & Zwitserlood (2003); Van Berkum, Koornneef, Otten & Nieuwland (2007) correspond to the vmPFC response. Their speculation follows from the work by Nieuwland et al. (2007), which supports the involvement of the vmPFC in language processing by evidence from referential ambiguity, which has been found to elicit greater vmPFC activation than referentially failing expressions. In ERP studies, comparable manipulations have elicited anterior negativities distinct from the N400 typically associated with semantic anomalies (Nieuwland & Van Berkum, 2008; Van Berkum et al., 2003, 2007). These anterior effects in ERP studies may therefore correspond to the AMF response observed in MEG. Since the current chapter aims to identify an ERP component that is sensitive to a pragmatic/linguistic violation, the next subsection reviews one of the studies on AMF effect in ERP studies, i.e., Van Berkum, Brown, Hagoort & Zwitserlood (2003).

2.4 AMF effect in ERP studies

Van Berkum, Brown, Hagoort & Zwitserlood (2003) investigated referential processing in spoken language by using ERPs to selectively track the neural mechanisms that establish reference during comprehension. In their ERP experiment, participants listened to stories in which a noun phrase such as *the girl* either uniquely referred to a single discourse referent as in (7a) or ambiguously referred to two equally plausible referents as in (7b).

- (7) a. Unambiguous:
David had told the boy and the girl to clean up their room before lunchtime. But the boy had stayed in bed all morning, and the girl had been on the phone all the time. David told **the girl** that had been on the phone to hang up.
- b. Ambiguous:
David had told the two girls to clean up their room before lunchtime. But one of the girls had stayed in bed all morning, and the other had been on the phone all the time. David told **the girl** that had been on the phone to hang up.

Referential ambiguity (e.g., *the girl*) when two girls had been introduced as in (7b) elicited a sustained negative ERP shift in the frontal field, within the time-window of 300–400 ms after onset, as depicted in Figure 4.

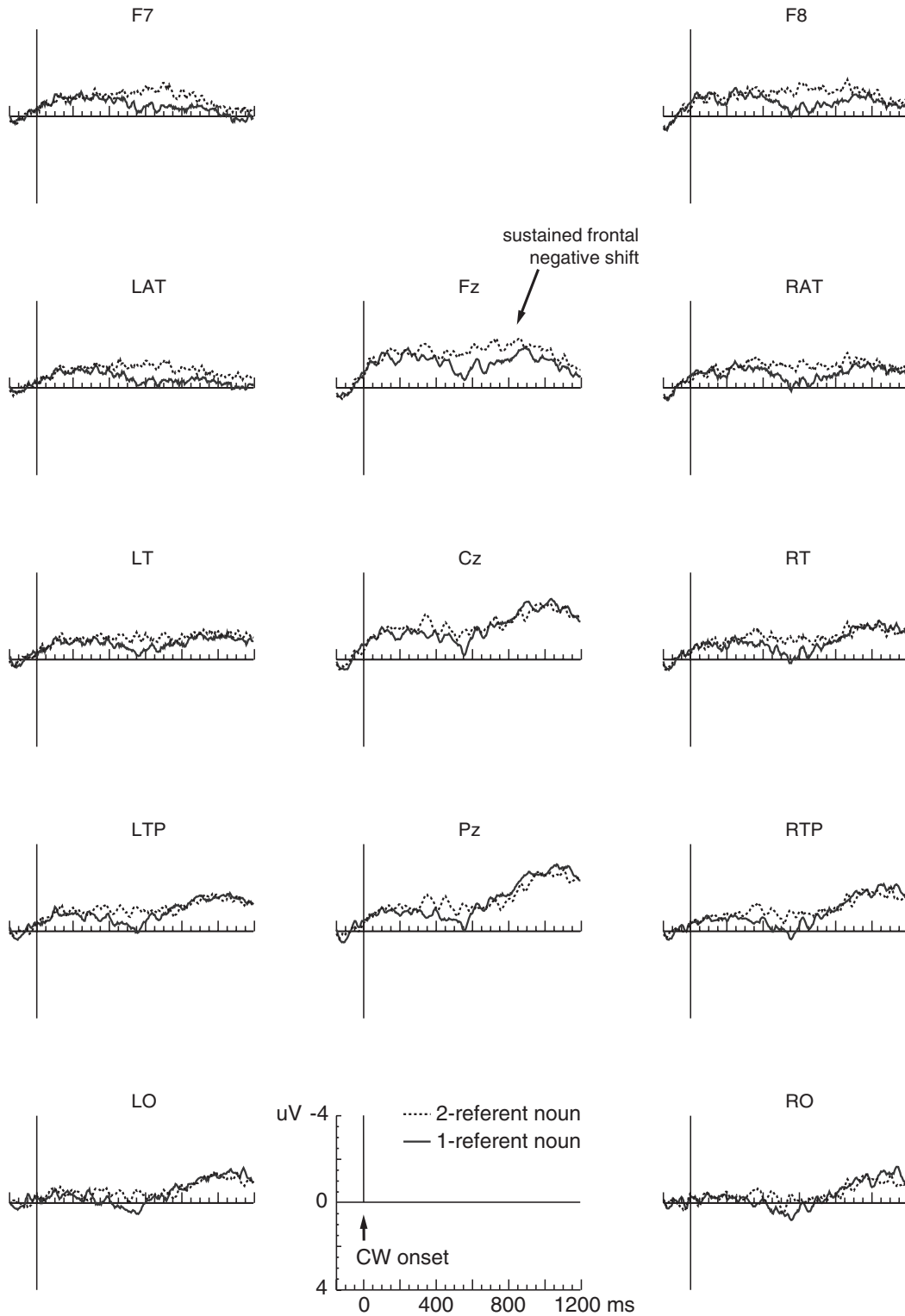


Figure 4: Van Berkum et al's (2003) sustained frontal negative shift (239)

Although discourse-induced referential ambiguity like the one in (7b) might appear to involve semantic computation, the ERP effect it elicits is distinct from the classic N400 typically associated

with world knowledge (in the sense of Pykkänen et al. (2009) ; “semantic” in Kutas & Hillyard (1980); Hagoort et al. (1998)) anomalies. The standard language-related N400 emerges around 150–200 ms after word onset, peaks near 400 ms, and is maximal at centro-parietal sites. By contrast, the referentially induced ERP effect is smaller in amplitude, has a later onset (around the N400 peak), shows an anterior distribution, and lacks a clear peak. Thus, difficulties with reference and difficulties with world knowledge give rise to qualitatively different ERP responses. Van Berkum, Koornneef, Otten & Nieuwland (2007); Nieuwland & Van Berkum (2008) also report that the same sustained frontal negative ERP component, which is qualitatively different from the classic N400 effect, is elicited when the subjects find difficulties in resolving discourse reference.³

2.5 Interim Summary

The following table summarizes the ERP and MEG effects that are elicited by the world knowledge violation (in the sense of Pykkänen et al. (2009)) and purely linguistic violations that are irreducible to world knowledge violations:

	ERP			MEG		
anomaly types	name	region	peak	name	region	time (ms)
world knowledge	N400	centro-parietal	400 ms	LIPC	left inferior	300–330
purely linguistic (sem/prag)	sustained negative shift	frontal	no clear peak	vmPFC	anterior midline	290–310

3 Research Questions

The primary objective of this chapter is to delineate the neurological distinctions between intuitions grounded in purely linguistic/pragmatic processes and those based on general world knowledge. In doing so, the discussion aims to identify ERP correlates of linguistic pragmatic/processing, in a manner parallel to the MEG evidence presented by Pykkänen et al. (2009) for semantic composition. In other words, the discussion in this chapter may be framed in terms of the following research questions:

- (8) Research questions
 - a. Can we dissociate world knowledge and purely linguistic violations which cannot be reduced to world knowledge in EEG experiments?
 - b. Can we identify an ERP component that signifies purely pragmatic processing? If so, is it sustained frontal negative shift (Van Berkum et al., 2003, 2007; Nieuwland & Van Berkum, 2008)?

In order to answer these questions, the chapter reexamines the finding of EEG experiment on Japanese evidentiality and modality reported in Hara et al. (2020), with the goal of situating these results within the broader inquiry into the neural basis of pragmatic computation.

³In Van Berkum et al. (2007); Nieuwland & Van Berkum (2008), this sustained frontal negative shift is coined “Nref effect” as it is insofar understood as a correlate of ambiguity of referents. In this chapter, I refrain from using this term for two reasons. First, it is better to give objective names to ERP components as kinds of language processing and ERPs are not in one-to-one correspondence. Second, as with Pykkänen et al. (2009), I speculate that this sustained frontal negative shift might be a neural response of purely linguistic/semantic/pragmatic violation that is comparable to the AMF effect, which does not necessarily involve discourse referent resolution.

4 Semantics and Pragmatics of Modality and Evidentiality

Evidentials refer to morphemes that indicate the source of the information on which a speaker bases their statement (Aikhenvald, 2004; Willett, 1988). These morphemes vary according to the type of evidence they convey. Willett (1988), for example, proposes a classification of such evidence types.

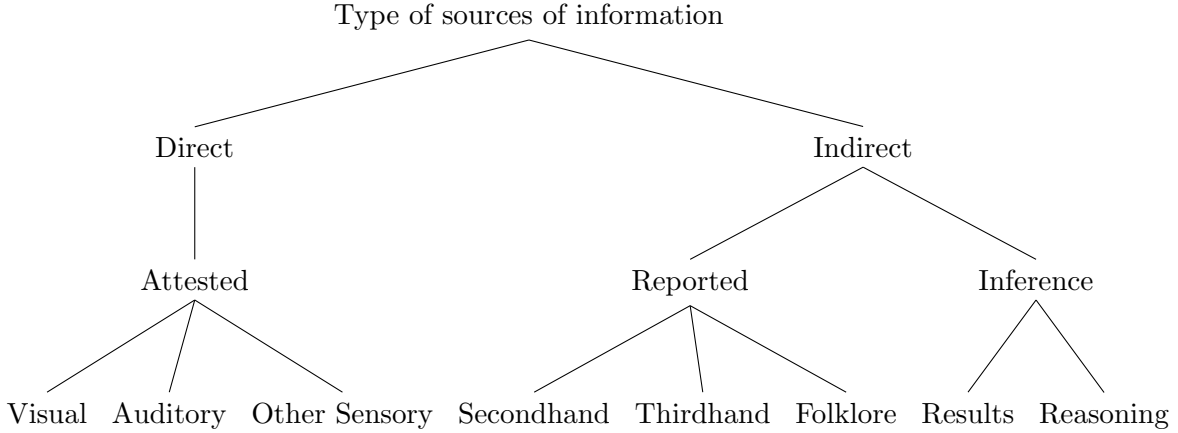


Figure 5: Willett’s (1988, 57) typology of evidence types

Japanese possesses a wide array of indirect evidential markers, including *yooda/mitaida/rasii* (‘it seems/appears’), *TP+sooda* (‘I hear’), and *V+sooda* (‘looks like’) (Aoki, 1986; McCready & Ogata, 2007; Hara, 2017). To the best of the author’s knowledge, McCready & Ogata (2007) were the first to provide a formal analysis of Japanese indirect evidentials. They argue that evidential markers should be understood as a type of epistemic modal. Much of the influential research on evidentiality in other languages (e.g., Izvorski, 1997; Matthewson et al., 2006) similarly treats evidentials within the broader category of epistemic modality. However, canonical modals in English, such as *must*, have been analyzed as belonging to inferential evidentials (von Fintel & Gillies, 2010; Degen et al., 2019), H&D argue that evidentials form a category separate from epistemic modality.

Specifically, *p-daroo* commits the speaker to the possibility of *p* being true through quantification over possible worlds. In (9), the speaker is weakly committed to the truth of the proposition ‘the streets are wet.’ In other words, the speaker asserts that there is at least one world among their doxastic worlds where the streets are wet.

- (9) Michi-ga nureteiru daroo.
 streets-NOM wet I-bet
 ‘The streets are wet, I bet/probably.’

On the other hand, *p-yooda* presupposes that the event *p* usually causes a state *q* and asserts that the speaker perceives *q*. In (10), the speaker assumes that the causal relation that rain causes wet streets is part of the world knowledge, hence in the common ground. Then, the speaker asserts that they perceive the cause, for example, wet streets.

- (10) Ame-ga futta yooda.
 rain-NOM fell EVID
 ‘It seems that it rained.’

This analysis is motivated by introspection-based intuitions in terms of the presence or absence

of epistemic commitment (Section 4.1) and whether or not the causal relation is encoded in the lexical semantics (Section 4.2).

4.1 Presence/Absence of Epistemic Commitment

H&D demonstrate that *yooda* differs from *daroo* in that the prejacent *p* in *p-yooda* does not constitute an at-issue commitment of the utterance, as it can be cancelled. As shown in (11) and (12), both a bare assertion *p* and *p-daroo* entail the speaker’s commitment to *p*, making cancellation impossible.

- (11) #Ame-ga futta kedo jitsu-wa futtenai.
rain-NOM fell but actually fall-NEG
‘#It rained but in fact it didn’t.’
- (12) #Ame-ga futta daroo kedo jitsu-wa futtenai.
rain-NOM fell I-bet but actually fall-NEG
‘#Probably, it rained but in fact it didn’t.’

By contrast, the prejacent *p* in *p-yooda* can be cancelled, as illustrated in (13):

- (13) Ame-ga futta yooda kedo, jitsu-wa futte-nai. (Dereka-ga mizu-o maitanda.)
rain-NOM fell EVID but actually fall-NEG someone-NOM water-ACC sprayed
‘It seems that it rained, but in fact it didn’t. (Someone sprayed water.)’

To summarize, if *p-yooda* were a type of modality, it would be expected to entail at least a weak commitment to *p*. However, because its prejacent is cancellable, *yooda* must be classified as a distinct category.

4.2 Causality in semantics or pragmatics?

H&D further argue that the concept of evidentiality must capture *asymmetric* causal relations: For instance, rain leads to wet streets, but not *vice versa*. McCready & Ogata (2007) (M&O, henceforth) develop a Bayesian modal semantics for several evidential expressions. Within M&O’s framework, evidence for asserting *p-yooda* is defined as some piece of information *q* such that learning *q* increases *a*’s subjective probability of *p*. In (14), for example, *a* comes to know that the streets are wet, which in turn raises *a*’s subjective probability of *p*; therefore, the use of *yooda* is licensed.

- (14) (Looking at wet streets)
Ame-ga futta yooda.
rain-NOM fell EVID
‘It seems that it rained.’

However, H&D demonstrate that the analysis yields incorrect predictions when *p* and *q* are reversed, as in (15). Acquiring the information that it is raining should likewise increase the agent’s subjective probability of “the streets are wet.” Hence, M&O incorrectly predict that the use of *yooda* is felicitous in (15).

- (15) (Looking at falling raindrops)
#Michi-ga nureteiru yooda.
streets-NOM wet EVID
‘#It seems that the streets are wet.’

From this observation, H&D argue that the evidence relevant for asserting *p-yooda* is some piece of information *q* such that *p* brings about *q*.

Daroo appears to be in complementary distribution with *yooda*. In particular, *p-daroo* is unacceptable when the speaker encounters information *q* such that *p* causes *q*, as in (16a), but it is acceptable when the information *q* is such that *q* leads to *p*, as in (16b).

- (16) a. (Looking at wet streets)
 #Ame-ga futta daroo.
 rain-NOM fell I-bet
 ‘#Probably, it rained.’
 b. (Looking at falling raindrops)
 Michi-ga nureteiru daroo.
 streets-NOM wet I-bet
 ‘Probably, the streets are wet.’

The current chapter follows Hara & Davis (2013) in assuming that, unlike *yooda*, causality is not lexically encoded in the semantics of *daroo*. Instead, the contrast described above results from pragmatic competition. *Yooda* carries stronger semantics in that it presupposes a more specific context than *daroo*. Since *yooda* provides a better fit for the context in (16a), *daroo* becomes infelicitous there. For further discussion, see Hara & Davis (2013).

In short, the semantics of *p-yooda* crucially depends on causality. More precisely, the *p* in *p-yooda* designates a cause event within a causal relation, where *p* causes *q*. By contrast, the apparent causal constraint on the use of *daroo* arises from pragmatic competition rather than from its lexical semantics.

4.3 Lexical semantics of *yooda* and *daroo*

Building on H&D’s observation, Hara (2017) develops a formulation of the semantics of *yooda* within Kaufmann’s (2013) Causal Premise Semantics, which extends Kratzer’s (2005) premise semantics by incorporating causal networks. Let **f** denote the premise background, **g** the ordering source, and *w* a possible world. The premise structure $\text{Prem}((\mathbf{f}^*\mathbf{g})(w))$ is derived by ranking Kratzer’s premise sets.⁴ The basic modal operators, *Must* and *May*, are defined as follows:

- (17) a. *Must*(*p*) is true at **f**, **g**, *w* iff *p* is a necessity relative to $\text{Prem}((\mathbf{f}^*\mathbf{g})(w))$.
 b. *May*(*p*) is true at **f**, **g**, *w* iff *p* is a possibility relative to $\text{Prem}((\mathbf{f}^*\mathbf{g})(w))$.

Hara et al. (2020) suggest that *daroo* functions as a necessity modal, which can be interpreted as follows:⁵

- (18) **Interpretation of *daroo*:** *Daroo*(*p*) is true at **f**, **g**, *w* iff *Must*(*p*) is true at **f**, **g**, *w*.

Turning to the evidential *yooda*, Hara’s (2017) account of causality is grounded in the interpretation of conditionals. Within the present framework, a causal premise background **f_c**[*p*] is derived by hypothetically updating the causal premise background **f_c** with the antecedent proposition *p*, as shown in (19).

- (19) **Hypothetical update:** For all *w*: **f**[*p*](*w*) := $\{\{p\}\}^*\mathbf{f}(w)$.

⁴For reasons of space, many definitions are omitted. See Hara (2017) and Kaufmann (2013) for details.

⁵*Daroo* also exhibits certain lexical restrictions that are not directly relevant to the present discussion. See Hara (2006, 2023).

Accordingly, the statement ‘ p causes q ’ is represented as $\text{Must}_p(q)$ in (20):

(20) **Causal rule:** $\text{Must}_p(q)$ is true at $\mathbf{f}_c, \mathbf{g}, w$ iff $\text{Must}(q)$ is true at $\mathbf{f}_c[p], \mathbf{g}, w$.

Finally, the interpretation of *yooda* can be defined as follows. As shown by H&D, asserting *p-yooda* does not commit the speaker to p . Accordingly, p contributes to the *antecedent* of the conditional rather than the consequent, i.e., $\text{Must}_p(q)$. In Hara’s (2017) original formulation, the causal component $\text{Must}_p(q)$ was part of the asserted content. In the present analysis, however, the causal component is placed in the presupposition, since it belongs to the common ground and is not asserted (for example, in (10), the speaker is not asserting that ‘rain causes wet streets’ but rather taking it as shared world knowledge). Thus, the interpretation of *yooda* is defined as in (21): $\text{Yooda}(p)$ presupposes the existence of some state q such that q is necessary relative to $\text{Prem}((\mathbf{f}_c[p] * \mathbf{g})(w))$, and asserts that the speaker perceives q at w .

(21) **Interpretation of *yooda*:**

$\text{Yooda}(p)$ is defined at $\mathbf{f}_c, \mathbf{g}, w$ iff $\exists q$ such that $\text{Must}_p(q)$ is true at $\mathbf{f}_c, \mathbf{g}, w$.

(presupposition)

If defined, $\text{Yooda}(p)$ is true at w iff the speaker perceives q at w .

(assertion)

As shown in (21), the assertion of *p-yooda* is evaluated solely at the actual/utterance world w . Following H&D, definition (21) correctly accounts for the interpretations and distributions of *yooda* presented above. In (13), the speaker presupposes that rain causes wet streets, while asserting only that wet streets were observed—an outcome typically caused by rain but also potentially by other factors, such as water being sprayed with a hose. Example (15) results in a presupposition failure, since it is part of the world knowledge that wet streets do not cause rain.

In summary, *yooda* belongs to the category of evidentiality, which is distinct from modality markers such as *daroo*. The semantics of *p-yooda* carries the presupposition that there exists an event q such that p causes q . The assertion of *p-yooda* does not commit the speaker to p , but merely conveys that the speaker perceives q . Accordingly, when a *yooda*-sentence presupposes an implausible causal relation as in (14), the utterance becomes anomalous due to a presupposition failure. In contrast, *p-daroo* is an epistemic necessity modal whose semantics involves quantification over possible worlds, and its assertion commits the speaker to p . Moreover, unlike *yooda*, *daroo* lacks a causality presupposition. As discussed in Section 4.2, the source of the anomaly of *p-daroo* in (16a) is not a presupposition failure from an implausible causal relation (i.e., wet streets cause rain), but rather a loss in the pragmatic competition. *Yooda* has stronger semantics, as it presupposes a more specific context than *daroo*. Because *yooda* aligns more appropriately with the context in (16a), the use of *daroo* is infelicitous in that case.

The EEG experiment reported in Hara et al. (2020) aims to provide empirical support for theoretical claims about the semantics of *yooda* and *daroo*. Specifically, *daroo* functions as a type of epistemic modality, whereas *yooda* does not. The semantics of *yooda* encodes an asymmetric causal relation, while the causal restriction associated with *daroo* arises pragmatically.

5 Reinterpreting Hara et al (2020)

Recall from Section 3 that this chapter reexamines Hara, Orita, Ying, Koshizuka & Sakai (2020) to answer the following questions:

(22) Research questions

- a. Can we dissociate world knowledge and purely linguistic violations which cannot be reduced to world knowledge in EEG experiments?
- b. Can we identify an ERP component that signifies purely pragmatic processing? If so, is it sustained frontal negative shift (Van Berkum et al., 2003, 2007; Nieuwland & Van Berkum, 2008)?

H&D’s introspection-based analysis discussed in the previous section concludes that the anomaly of *CEy* (23b) is a world knowledge violation because the *p-yooda* utterance forces the parser to infer an implausible causal relation, i.e., wet streets cause rain, while the anomaly of *ECd* (23d) is a pragmatic violation because the speaker fails to employ *ECy* (23a), which is optimal for the given context because the speaker observed indirect evidence, i.e., wet streets.⁶

- (23) a. *ECy*: Effect-Cause-*yooda* :
- Michi-ga nureteiru. Ame-ga futta yooda.
streets-NOM wet rain-NOM fell EVID
‘The streets are wet. It seems that it rained.’
- b. *CEy*: Cause-Effect-*yooda*:
- #Ame-ga futta. Michi-ga nureteiru yooda.
rain-NOM fell streets-NOM wet EVID
‘#It rained. It seems that the streets are wet.’
- c. *CEd*: Cause-Effect-*daroo*:
- Ame-ga futta. Michi-ga nureteiru daroo.
rain-NOM fell streets-NOM wet I-bet
‘It rained. The streets are wet, probably.’
- d. *ECd*: Effect-Cause-*daroo*:
- #Michi-ga nureteiru. Ame-ga futta daroo.
streets-NOM wet rain-NOM fell I-bet
‘The streets are wet. It rained, probably’

Thus, the working hypothesis is formulated as follows:

- (24) Hypothesis:
- The anomalous Cause-Effect-*yooda* sentence is a world knowledge violation, while the anomalous Effect-Cause-*daroo* is a purely linguistic/pragmatic violation.

Accordingly, the hypothesis makes the following predictions:

- (25) Predictions:
- a. The anomalous Cause-Effect-*yooda* and the anomalous Effect-Cause-*daroo* yield different neural responses.
 - b. The anomalous Cause-Effect-*yooda* and the anomalous Effect-Cause-*daroo* yield N400 and AMF (sustained frontal negative shift) effect, respectively.

The results, stated at the outset, are as follows: First, Hara et al. (2020) interpret the anterior negativity observed in the *ECy* condition as a Left-Anterior Negativity (LAN), which has been proposed as a neural correlate of successful causal inference (Baggio et al., 2008; Cohn & Kutas, 2015). Second, as expected, anomalous *yooda* sentences elicited an N400 effect. In contrast, no

⁶In the original discussion of Hara et al. (2020), *CEy* (23b) was considered as a semantic violation.

significant difference was observed between control and target *daroo* sentences. Thus, although the study revealed a distinction between the processing of world knowledge and pragmatic violations, it did not succeed in isolating a neural correlate specific to pragmatic anomaly. Instead, relative to *yooda* sentences (both control and target), *daroo* sentences elicited an N400 effect previously associated with the processing of modality, that is, the evaluation of multiple possible worlds (Dwivedi et al., 2006; Kulakova et al., 2014; Kulakova & Nieuwland, 2016). It is plausible that this modality-induced N400 masked the AMF effect (sustained frontal negative shift), which has been proposed to manifest as a purely linguistic processing.

5.1 LAN as Successful Causal Inference

The final auxiliary *yooda* in the control ECy condition elicited a significantly greater negative deflection at the left anterior region compared to the anomalous CEy condition; see Figure 6).

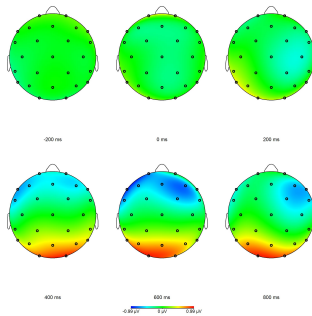


Figure 6: ECy–CEy

Hara et al. (2020) interpret the anterior negativity elicited by the felicitous ECy condition (Figure 7) as a Left-Anterior Negativity (LAN), which has been argued to serve as a neural correlate of successful causal inference (Baggio et al., 2008; Cohn & Kutas, 2015).⁷

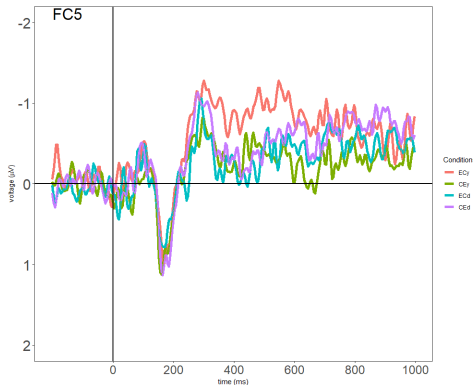


Figure 7: LAN (elicited by ECy) or P600 (elicited by CEy)

⁷In Baggio et al. (2008); Cohn & Kutas (2015), causal inferences are triggered by the semantic content of the stimuli and background world knowledge, rather than by a specific lexical item that overtly encodes causality, such as *yooda*. Accordingly, Hara et al. (2020) assume that LAN indexes successful causal inference in general, without differentiating whether the causal relation arises from semantic content or from a lexicalized causal marker.

5.2 World Knowledge vs. Pragmatic violations

At the posterior region, the anomalous *CEy* condition elicited a significantly greater negative deflection than the control *ECy* condition. The posterior negativity observed in Figure 8 is regarded as N400, a neural correlate of world knowledge violation.

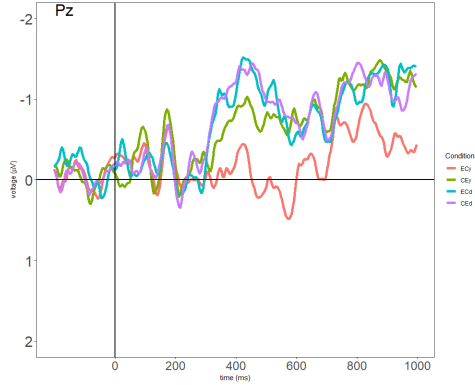


Figure 8: N400 (elicited by *CEy*)

While the two *yooda* conditions exhibited significant differences, no significant difference was observed between the two *daroo* conditions, *CED* and *ECd*. One possibility is that the modality-induced N400 discussed in the next subsection concealed the AMF effect or the sustained frontal negative shift.⁸

5.3 Modality-induced N400

Figure 9 indicates that all (both felicitous and anomalous) *daroo*-sentences elicited more pronounced negative deflections relative to all *yooda* sentences in both the central and posterior regions.

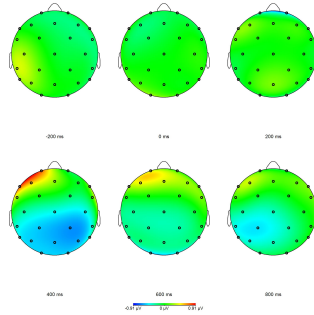


Figure 9: *daroo*—*yooda*

Hara et al. (2020) argue that the posterior and central negativity observed for *daroo* (Figure 10) corresponds to a modal-induced N400 effect, reflecting the processing demands associated with modality. Specifically, the interpretation of modality requires considering multiple possible worlds,

⁸Hara et al. (2020) attribute the absence of difference among *daroo* sentences to the nature of pragmatic anomaly. Specifically, the *ECd* condition is only pragmatically anomalous, given that a more appropriate auxiliary, namely *yooda*, is available for the Effect–Cause context. Such pragmatic anomaly, however, appears insufficient to elicit measurable effects on the waveforms. By contrast, the *CEy* condition constitutes a semantic anomaly, since the semantics of *yooda* requires attachment to the cause event within a cause–effect relation.

thereby increasing cognitive load.⁹ This interpretation aligns with previous findings (Dwivedi et al., 2006; Kulakova et al., 2014; Kulakova & Nieuwland, 2016).¹⁰

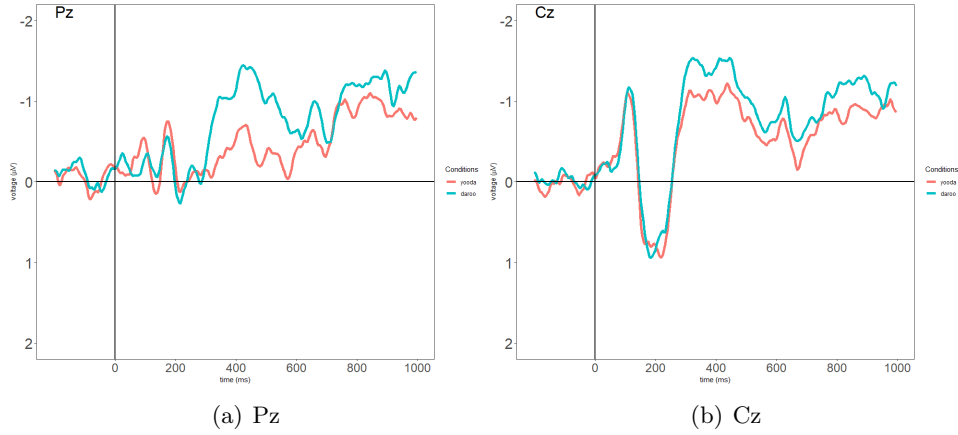


Figure 10: N400 elicited by *daroo*

6 Conclusion

The meaning of a sentence can be anomalous in different ways. First, the sentence is strange because it describes an implausible or impossible situation (world knowledge violation). Second, the sentence is ill-formed because it does not obey the rules of the semantic composition (semantic violation). Third, the sentence is infelicitous because it is not the optimal choice for the given context (pragmatic violation). The MEG study by Pyllkänen, Oliveri & Smart (2009) successfully identified vmPFC/AMF and LIPC effects as semantic and world knowledge violations, respectively. It is yet to be examined whether the differences between the world knowledge, semantic, and pragmatic violations correspond to the EEG responses. Incidentally, evidence for the involvement of the vmPFC in comprehension comes from studies on referential ambiguity, which elicits greater vmPFC activity than referential failure (Nieuwland et al., 2007). In ERPs, the EEG studies by Van Berkum et al. (2003, 2007); Nieuwland & Van Berkum (2008) show that sustained frontal negative shift is a neural index of difficulties in discourse referent resolutions. Thus, it is highly likely that these sustained frontal negative responses in ERPs correspond to the vmPFC/AMF effect in MEG studies.

To conclude, it is useful to revisit the research questions that guided this study:

(26) Research questions

- a. Can we dissociate world knowledge and purely linguistic violations which cannot be reduced to world knowledge in EEG experiments?
- b. Can we identify an ERP component that signifies purely pragmatic processing? If so, is it sustained frontal negative shift (Van Berkum et al., 2003, 2007; Nieuwland & Van Berkum, 2008)?

⁹The causal meaning ‘ p causes q ’ of p -*yooda* is expressed as the modal statement **Must** $p(q)$, but this is encoded in the presupposition rather than the assertional content, which refers only to the actual world $w@$.

¹⁰Importantly, the observed N400 cannot be attributed to lexical infrequency effects (c.f. Van Petten & Kutas, 1990), since, as shown in Hara et al. (2020), *daroo* occurs more frequently than *yooda* in the BCCWJ (30,686 instances of *yooda* and 47,538 instances of *daroo*).

The contrast between Japanese evidential *yooda* and modal *daroo* provides an ideal testing ground for addressing the questions outlined above. The introspection-based analyses by Davis & Hara (2014); Hara & Davis (2013); Hara (2017) demonstrate that *yooda* encodes causality in its lexical semantics and an anomalous *yooda* sentence gives rise to a world knowledge violation, while *daroo* is a simple modal without causal presupposition and an anomalous *daroo* sentence leads to a pragmatic violation due to a loss in pragmatic competition.

The results of the EEG experiment by Hara et al. (2020) provide the answer to the first question in (26). As predicted, anomalous *yooda* sentences elicited a greater N400 effect. In contrast, no significant difference emerged between control and target *daroo* sentences. Thus, although the study demonstrated a distinction between the processing of world knowledge and pragmatic violations, it did not succeed in identifying a neural correlate specific to pragmatic anomaly, failing to answer the second question. Incidentally, relative to both control and target *yooda* sentences, *daroo* sentences elicited an N400 effect previously linked to the processing of modality—that is, the evaluation of multiple possible worlds (Dwivedi et al., 2006; Kulakova et al., 2014; Kulakova & Nieuwland, 2016). It is therefore plausible that this modality-induced N400 masked the AMF effect, which has been proposed to manifest as a sustained frontal negative shift in ERP studies.

For future research, the interference of modality-induced N400 may be circumvented by conducting EEG experiments with stimuli that exclude modality but involve pragmatic competition. For example, as discussed by Zeevat (2004), a sentence with English additive particle *too* is the optimal choice when the topic of the utterance is already addressed. Thus, (27b) without the particle is anomalous since it is not the best candidate in the pragmatic competition.

- (27) a. Bill will come tonight.
b. John will come #(. too).

Similarly, the contrast between Osaka Japanese *nen* and *de* discussed by Hara & Kinuhata (2012) is another good example of pragmatic competition. According to Hara & Kinuhata (2012), *nen* is a default assertion marker that publicize the speaker’s belief. In contrast, *de* is a particle that marks the relevant action for the addressee. Thus, when the speaker intends to simply publicize their belief, the use of *nen* is the optimal choice as in (28).

- (28) A: Nande kasa motten no?
why umbrella carry Q
‘Why are you carrying an umbrella?’
B: Ame furu-nen/#de.
rain fall-NEN/DE
‘It’s going to rain.’

On the other hand, in a context where the speaker wants to make some suggestion to the addressee, *de* is the optimal choice:

- (29) Kasa motte ike-ya! Ame furu-de/#nen.
umbrella carry go.IMP-PRT rain fall-DE/NEN
‘Take the umbrella! It’s going to rain-*de*.’

Lastly, the distribution of Japanese *ka/noka* questions seems to be determined by pragmatic blocking. In a nutshell, a *ka*-question is the default question which simply asks for the fact, while a *noka*-question has an extra function that intervenes the addressee’s intention (Ijima, 2010). Thus, it is very natural for a taxi driver to ask the *ka*-question (30a), while the *noka* question (30b) would sound rude. In contrast, in a situation where a businessperson suddenly stands up with some

documents and heads outside, and the colleague wonders where they are going, the colleague would utter (30b), but not (30a).

- (30) a. doko-e iki masu-ka?
 where-DAT go POL-Q
 ‘Where are you going?’ (KA)
- b. doko-e iku n(o)-desu-ka?
 where-DAT go NML-COP.POL-Q
 ‘Where are you going?’ (NOKA)
- (Yoshida, 1994, 1)

An examination of the neural responses to these contrasts may enable the identification of correlates of purely linguistic or pragmatic violations that cannot be reduced to violations of world knowledge.

References

- Aikhenvald, Alexandra Y. 2004. *Evidentiality*. Oxford: Oxford University Press.
- Aoki, Haruo. 1986. Evidentials in Japanese. In Wallace Chafe & Johanna Nichols (eds.), *Evidentiality: The Linguistic Coding of Epistemology*, Norwood, New Jersey: Ablex Publishing Corporation.
- Baggio, Giosuè, Michiel van Lambalgen & Peter Hagoort. 2008. Computing and recomputing discourse models: An ERP study. *Journal of Memory and Language* 59. 36–53.
- Blutner, Reinhard & Henk Zeevat (eds.). 2004. *Optimality Theory and Pragmatics*. New York: Palgrave Macmillan.
- Cohn, Neil & Marta Kutas. 2015. Getting a cue before getting a clue: Event-related potentials to inference in visual narrative comprehension. *Neuropsychologia* 77. 267–278. doi: 10.1016/j.neuropsychologia.2015.08.026.
- Covington, M. 1981. *Evidence Synthesis Guide 1: Understanding Evidence Synthesis*. Bloomington, IN: SAGE Publications Ltd. doi:10.4135/9781036222710.
- Davis, Christopher & Yurie Hara. 2014. Evidentiality as a causal relation: A case study from Japanese *youda*. In Christopher Piñón (ed.), *Empirical Issues in Syntax and Semantics* 10, .
- Degen, Judith, Andreas Trotzke, Gregory Scontras, Eva Wittenberg & Noah D. Goodman. 2019. Definitely, maybe: A new experimental paradigm for investigating the pragmatics of evidential devices across languages. *Journal of Pragmatics* 140. 33–48.
- Dwivedi, Veena D., Natalie A. Phillips, Maude Laguë-Beauvais & Shari R. Baum. 2006. An electrophysiological study of mood, modal context, and anaphora. *Brain research* 1117 1. 135–53.
- von Fintel, Kai & Anthony S. Gillies. 2010. Must . . . stay . . . strong! *Natural Language Semantics* 18(4). 351–383. doi:10.1007/s11050-010-9058-2.
- Hagoort, Peter, Lea Hald, Marcel Bastiaansen & Karl Petersson. 1998. Integration of word meaning and world knowledge in language comprehension. *Science* 304(5669). 438–441. doi: 10.1126/science.1095455.

- Hara, Yurie. 2006. *Japanese Discourse Items at Interfaces*. Newark, DE: University of Delaware dissertation.
- Hara, Yurie. 2017. Causality and evidentiality. In Alexandre Cremers, Thom van Gessel & Floris Roelofsen (eds.), *Proceedings of the 21st Amsterdam Colloquium*, 295–304.
- Hara, Yurie. 2023. **daroo ka*↑: the interplay of deictic modality, sentence type, prosody and tier of meaning. *Natural Language & Linguistic Theory* 42(1). 95–152. doi:10.1007/s11049-023-09573-6.
- Hara, Yurie & Christopher Davis. 2013. *Darou* as a deictic context shifter. In Kazuko Yatsushiro & Uli Sauerland (eds.), *Proceedings of Formal Approaches to Japanese Linguistics 6 (FAJL 6)* MIT Working Papers in Linguistics 66, 41–56.
- Hara, Yurie & Tomohide Kinuhata. 2012. Osaka Japanese *nen*: One-sided public belief and parat-actic association. *Sprache und Datenverarbeitung: International Journal for Language Data Processing* 49–70.
- Hara, Yurie, Naho Orita, Deng Ying, Takeshi Koshizuka & Hiromu Sakai. 2020. A neurolinguistic investigation into semantic differences of evidentiality and modality. In Nikola Kompa Mingya Liu Jutta L. Mueller Michael Franke & Juliane Schwab (eds.), *Proceedings of Sinn und Bedeutung*, vol. 24, 273–290.
- Horn, Laurence. 2002. Uncovering the un-word: A study in lexical pragmatics. *Sophia Linguistica* 49. 1–64.
- Ijima, Masahiro. 2010. Noda bun no kinoo to koozoo (the function and structure of noda sentences). *Nihongogaku Ronshuu* 6.
- Izvorski, Roumyana. 1997. The Present Perfect as an Epistemic Modal. *the proceedings of SALT* 7. 222–239.
- Kaufmann, Stefan. 2013. Causal premise semantics. *Cognitive Science* 37. 1136–1170.
- Kratzer, A. 2005. Constraining premise sets for counterfactuals. *Journal of Semantics* 22. 153–158.
- Kulakova, Eugenia, Dominik Freunberger & Dietmar Roehm. 2014. Marking the counterfactual: ERP evidence for pragmatic processing of German subjunctives. *Frontiers in human neuroscience* 8. 548. doi:10.3389/fnhum.2014.00548.
- Kulakova, Eugenia & Mante Nieuwland. 2016. Pragmatic skills predict online counterfactual comprehension: Evidence from the N400. *Cognitive Affective & Behavioral Neuroscience* in press. doi:10.3758/s13415-016-0433-4.
- Kutas, M. & S. A. Hillyard. 1980. Reading senseless sentences: Brain potentials reflect semantic incongruity. *Science* 203–205.
- Matthewson, Lisa, Hotze Rullmann & Henry Davis. 2006. Evidentials are epistemic modals in St’át’imcets. In Masaru Kiyota, James L. Thompson & Noriko Yamane-Tanaka (eds.), *Papers for the 41st International Conference on Salish and Neighbouring Languages* University of British Columbia Working Papers in Linguistics 18, 221–263.
- McCready, E. & Norry Ogata. 2007. Evidentiality, modality and probability. *Linguistics and Philosophy* 30(2). 35–63.

- Nieuwland, Mante, Karl Petersson & Jos Van Berkum. 2007. On sense and reference: Examining the functional neuroanatomy of referential processing. *NeuroImage* 37(3). 993–1004. doi: 10.1016/j.neuroimage.2007.05.048.
- Nieuwland, Mante & Jos Van Berkum. 2008. The interplay between semantic and referential aspects of anaphoric noun phrase resolution: Evidence from erps. *Brain and Language* 106(2). 119–131. doi:10.1016/j.bandl.2008.05.001.
- Pylkkänen, L & B Mcelree. 2007. An meg study of silent meaning. *Journal of Cognitive Neuroscience* 19. 1905–1921.
- Pylkkänen, Liina, Bridget Oliveri & Andrew J. Smart. 2009. Semantics vs. world knowledge in prefrontal cortex. *Language and Cognitive Processes* 24(9). 1313–1334. doi: 10.1080/01690960903120176. <https://doi.org/10.1080/01690960903120176>.
- Shamay-Tsoory, S, R Tomer & J Aharon-Peretz. 2005. The neuroanatomical basis of understanding sarcasm and its relationship to social cognition. *Neuropsychology* 19(3). 288–300. doi: 10.1037/0894-4105.19.3.288.
- Van Berkum, Jos, Colin Brown, Peter Hagoort & Pienie Zwitserlood. 2003. Event-related brain potentials reflect discourse-referential ambiguity in spoken language comprehension. *Psychophysiology* 40(2). 235–248. doi:10.1111/1469-8986.00025.
- Van Berkum, Jos, Arnout Koornneef, Marte Otten & Mante Nieuwland. 2007. Establishing reference in language comprehension: An electrophysiological perspective. *Brain Research* 1146. 158–171. doi:10.1016/j.brainres.2006.06.091.
- Van Petten, C. & M. Kutas. 1990. Interactions between sentence context and word frequency in event-related brain potentials. *Memory and Cognition* 18(4). 380–393. doi: doi:10.3758/bf03197127.
- Willett, T. 1988. A cross-linguistic survey of the grammaticalization of evidentiality. *Studies in Language* 12(1).
- Yoshida, Shigeaki. 1994. Gimonbun no shoruiki to sono jitsugen keishiki: *nodesuka/masuka* gata gimonbun no yohoo o megutte (a taxonomy of questions and their realized forms: the usage of *nodesuka/masuka* questions)). *Shimadai Kokubun* 22. 1–13.
- Zeevat, Henk. 2004. Particles: Presupposition triggers, context markers or speech act markers. In *Optimality Theory and Pragmatics*, 91–111. Palgrave MacMillan.